

RESEARCH PAPER

Reductions in leaf area index, pod production, seed size, and harvest index drive yield loss to high temperatures in soybean

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Abstract

Improvements in genetics, technology, and agricultural intensification have increased soybean yields; however, adverse climate conditions may prevent these gains from being fully realized in the future. Higher growing season temperatures reduce soybean yields in key production regions including the US Midwest, and better understanding of the developmental and physiological mechanisms that constrain soybean yield under high temperature conditions is needed. This study tested the response of two soybean cultivars to four elevated temperature treatments (+1.7, +2.6, +3.6, and +4.8 °C) in the field over three growing seasons and identified threshold temperatures for response and linear versus non-linear trait responses to temperature. Yield declined non-linearly to temperature, with decreases apparent when canopy temperature exceeded 20.9 °C for the locally adapted cultivar and 22.7 °C for a cultivar adapted to more southern locations. While stem node number increased with increasing temperature, leaf area index decreased substantially. Pod production, seed size, and harvest index significantly decreased with increasing temperature. The seasonal average temperature of even the mildest treatment exceeded the threshold temperatures for yield loss, emphasizing the importance of improving temperature tolerance in soybean germplasm with intensifying climate change.

Keywords: Climate change, *Glycine max*, harvest index, leaf area index, temperature stress, yield.

Introduction

Soybean (*Glycine max* Merr.) is an important crop to global food supply, providing a major source of plant protein and oil (Ainsworth *et al.*, 2012). The USA is one of the world's leading producers of soybean and accounts for ~25% of global production (USDA National Agricultural Statistics Survey, 2020). Global soybean yields have increased from 1.1 t ha⁻¹ to 2.8 t ha⁻¹ over the past 60 years due to improvements in genetics, technology, and agricultural intensification (Lobell and Gourdji, 2012; Mueller *et al.*, 2016; FAO, 2020). Soybean yields are expected to continue to rise as genetics and technology improve further; however, climate conditions detrimental to crop productivity could prevent these gains from being fully realized. Studies of historic yield and climate data indicate that increasing growing season temperatures have negatively affected yields (Mourtzinis *et al.*, 2015; Kukul and Irmak, 2018; Eck *et al.*, 2020). Warming in the Midwest USA from 1994 to 2013 caused an estimated 11.3% net decrease in soybean yields (Mourtzinis *et al.*, 2015). Modeled predictions of crop response to further warming suggest that a warmer future climate will be further detrimental to yields (Zhao *et al.*, 2017; Wang *et al.*, 2020). However, there is considerable uncertainty in these predictions (Zhu *et al.*, 2021), and the trends are based on both spatial and temporal variation in growing season temperatures. Uncertainty also stems from use of past statistical correlations which may or may not hold outside of the range in which data were collected, and from an incomplete mechanistic understanding of the effects of high temperature stress on soybean (Iizumi *et al.*, 2009; Lobell and Burke, 2010; Rosenzweig *et al.*, 2014). Accurate predictions of soybean yield under future growing conditions require further experiments that measure high temperature effects on developmental and physiological processes in a realistic field environment.

Soybean yield is determined by the number of seeds per area and the mass of the soybean seeds. Historical analyses of cultivars released throughout the past 90 years indicate that a major driver of the increase in yields is greater seed number per plant, with less change in individual seed mass (Morrison *et al.*, 2000; Jin *et al.*, 2010; Rincker *et al.*, 2014). However, high temperature stress can impact both seeds per area and the mass of individual seeds (Baker *et al.*, 1989; Gibson and Mullen, 1996; Heinemann *et al.*, 2006; Tacarindua *et al.*, 2012, 2013). Three determinate, maturity group (MG) IV soybean cultivars showed linear decreases in individual seed mass as average growing season temperature increased from ~26 °C to 30 °C, associated with decreased cell number and decreased seed growth rate (Tacarindua *et al.*, 2012). Above-ground biomass (AGB) and harvest index (HI) also linearly decreased in those cultivars across that temperature range (Tacarindua *et al.*, 2013). However, other studies have found that soybean yield was not affected by temperature increases from 26 °C to 39 °C (Baker *et al.*, 1989). Furthermore, soybeans of different MGs respond differently to warming treatments (Kumagai and Sameshima, 2014). Late maturing Japanese cultivars

increased maximum leaf area, pod and seed number, and seed yield when temperatures were increased by 4.8 °C to 5.7 °C above ambient; however, yield was unchanged by temperature in an early maturing cultivar (Kumagai and Sameshima, 2014). In the USA, soybean cultivars are classified into 13 MGs (000–10) based on the number of days from planting to maturity (Boerma and Specht, 2004), with additional gradations within MGs denoted by a decimal following the MG number. Late-maturing cultivars are adapted for lower latitude growing sites with shorter photoperiods and warmer temperatures. Changes in climate, genetics, and agricultural practices have caused MG zones in the USA to shift north over time (Zhang *et al.*, 2007; Mourtzinis and Conley, 2017). A series of trials comparing multiple MGs simultaneously at various latitudes across the USA determined that Champaign, Illinois, where our study was located, is within the MG 3.5–4.0 zone, while southern Illinois is within the MG 4.0–4.5 zone (Mourtzinis and Conley, 2017). Recent surveys have also suggested that the MG 4 zone may be wider than originally thought as changes in farming practices, including earlier planting dates, favor late maturing MG 4 cultivars (Zhang *et al.*, 2007). Rising growing season temperatures may also cause the MG 4 zone to expand into central Illinois as the climate becomes less favorable for MG 3. The variable experimental results described above indicate that broad generalizations of the temperature response of soybean across different MGs and regions may not be possible, and studies of adapted and future germplasm in major growing regions are still needed to fully understand soybean response to rising temperature.

Lower yields in soybeans grown at elevated temperature could be caused by altered development, canopy size, and/or length of developmental periods. Less stem and leaf tissue under high temperatures would reduce the leaf area per unit ground area. Reduced leaf area reduces the amount of solar energy that can be intercepted and used in photosynthesis for carbon assimilation (Monteith, 1977; Welles and Norman, 1991). Also, soybean yield is determined in large part by the number of stem nodes available for flower and pod formation; fewer nodes contribute to fewer flowers and pods (Kahlon *et al.*, 2011). Both leaf and stem node development can be impaired by high growing season temperatures (Tenorio *et al.*, 2017). Additionally, high temperatures can directly impact flower, pod, and seed development as these processes are highly sensitive to heat stress (van Schaik and Probst, 1958; Mann and Jaworski, 1970; Djanaguiraman *et al.*, 2013). Reproductive development is reportedly more sensitive to heat stress than vegetative development; studies have shown that as little as 3 d of exposure to high temperatures during critical periods can reduce yield (Egli and Wardlaw, 1980; Sage *et al.*, 2015; Siebers *et al.*, 2015; Thomey *et al.*, 2019).

The numerous soybean traits affected by temperature make it difficult to predict and accurately model yield responses

to high growing temperatures. Statistical models have demonstrated historic effects of high temperatures on yield, but these models are less reliable when predicting responses to growing temperatures outside the range represented in calibration data (Iizumi *et al.*, 2009). Better understanding of the process(es) through which heat stress reduces yield will improve the accuracy of yield predictions under climate change scenarios and indicate desirable traits to improve yields (Lobell and Burke, 2010; Rosenzweig *et al.*, 2014; Mistry *et al.*, 2017), but much of our process-level understanding comes from controlled environments with stable day/night temperature treatments, increased humidity, and restricted root growth (McConnaughay *et al.*, 1993; Mitchell *et al.*, 1993; Wilhelm *et al.*, 1999). Natural field conditions present much more variable environments, which can potentially change temperature responses (Ruiz-Vera *et al.*, 2013; Kronenberg *et al.*, 2020). Therefore, this study investigated soybean yield components under a range of increasing growing temperatures in the field using a temperature free-air controlled enhancement (T-FACE) system (Kimball *et al.*, 2008; Eyshi Rezaei *et al.*, 2015). The system uses infrared heaters deployed above a crop canopy, which provide a similar effect to solar heating of leaves (Kimball *et al.*, 2008). Previous open-air studies of soybean have observed the combined effects of elevated [CO₂] and air temperature (Ruiz-Vera *et al.*, 2013; Köhler *et al.*, 2017). This method enabled observations of the combined effects of two future climate change conditions (elevated [CO₂] and temperature) but placed practical constraints on the range of high temperature conditions that could be observed, as space, heating equipment, and power supplies were limited. Our current study investigated soybean grown under a gradient of ele-

vated temperatures and tested for thresholds above which processes declined to provide response-surface information for process-based models. Based on previous modeling studies (Schlenker and Roberts, 2009), we hypothesized that the soybean yields would respond non-linearly to increasing temperature, with accelerated development and damage to reproductive structures contributing to yield loss.

Materials and methods

Experimental design and heating treatments

In the summers of 2017–2019, two indeterminate, commercial cultivars of soybean (*Glycine max*) from MGs III and IV were grown under five target temperature conditions (ambient, +1.5, +3.0, +4.5, and +6.0 °C) at the SoyFACE research facility in Champaign, Illinois (Supplementary Fig. S1). Local temperature and precipitation data for the study period (2017–2019) as well as historic normals (1981–2010) were obtained from a monitoring station at Champaign Urbana Willard Airport located ~3 km from SoyFACE (NOAA/NCEI, Table 1). The MG III cultivar (MG 3.6) is adapted to central Illinois where SoyFACE is located, while the MG IV cultivar (MG 4.2) is a later maturing cultivar better suited to southern Illinois. Soybeans were planted at a high population density of 431 400 seeds per hectare in 38 cm rows on 29 May 2017, 17 May 2018, and 31 May 2019. Soybeans were grown in annual rotation with maize and were not fertilized. Pre-emergent herbicides (atrazine, saflufenacil, and dimethenamid-P) were used each year, with hand-weeding done within the experimental plots as necessary. In 2017 and 2018, there was one heating array set to each of the four elevated temperature treatments (Supplementary Fig. S1), and in 2019 there were two arrays set to each of the four temperature treatments. In 2019, trait values from duplicate heating treatments were averaged before analysis. Heating treatments began on 14 June 2017, 18 June 2018, and 11 July, 2019 after canopy closure, and remained on throughout the growing seasons until plants reached maturity (Fig. 1). Ambient temperature control plots were adjacent to the heated plots (Supplementary Fig. S1).

Table 1. Duration of reproductive developmental stages (in days) in ambient and elevated temperature plots in 2017, 2018, and 2019

	R1 (Begin bloom)	R2 (Full bloom)	R3 (Begin pod)	R4 (Full pod)	R5 (Begin seed)	R6 (Full seed)
2017						
Ambient	11	19	9	5	11	NA
+1.7 °C	9	23	4	9	10	NA
+2.6 °C	10	22	7	7	10	NA
+3.6 °C	10	24	9	5	9	NA
+4.8 °C	10	25	9	4	10	NA
2018						
Ambient	8	12	13	7	22	22
+1.7 °C	7	13	16	7	21	23
+2.6 °C	8	20	12	5	20	27
+3.6 °C	6	26	10	7	13	NA
+4.8 °C	8	27	9	11	11	31
2019						
Ambient	8	12	9	6	13	31
+1.7 °C	7	13	10	6	12	33
+2.6 °C	7	12	9	5	14	32
+3.6 °C	8	12	9	6	14	32
+4.8 °C	6	14	9	5	16	28

NA: data not measured or not available.

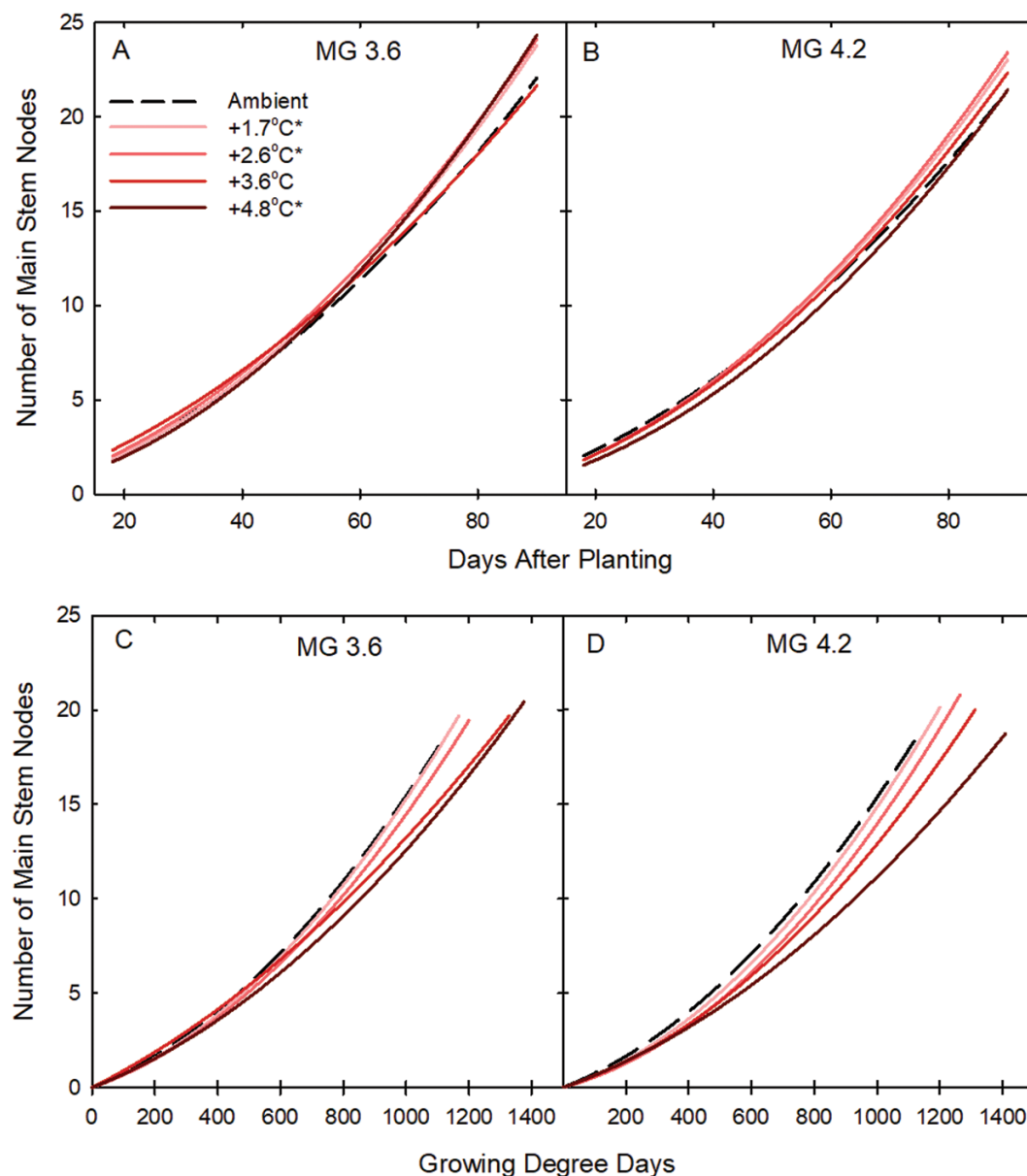


Fig. 1. Vegetative development in two cultivars of soybean (MG 3.6 and MG 4.2) averaged over 3 years of growth at ambient, +1.7, +2.6, +3.6, and +4.8 °C. Lines show the non-linear fits to the measurements of main stem nodes, and significant differences between the non-linear regressions for development at ambient and elevated temperature treatments are indicated with an asterisk (* $P < 0.05$).

Heating arrays consisted of six infrared heaters (Salamander Aluminum Extrusion Reflector Assembly Housing for Ceramic Infrared Heaters; Mor Electric Heating) each containing four infrared heating elements (Mor-FTE 1000 W, 240 V heaters; Mor Electric Heating Association). The arrays were placed ~1.2 m above the canopy in a hexagonal pattern and at 45° angles towards the center of the experimental plots. This design provided a uniform heated area of 7.1 m² (Ruiz-Vera *et al.*, 2013). Heat output was regulated by an industrial dimmer system (model LCED-2484, 240 V, 35 A; Kalglo Electronics). The dimmer circuit board controlled the output up to 24 000 W of infrared heating power. Ambient and elevated canopy temperatures were measured using infrared radiometers (SI-121; Apogee Instruments) connected to a datalogger (CR1000 Micrologger; Campbell Scientific). The datalogger was programmed with a proportional–integrative–derivative feed-

back control system to maintain the elevated temperature treatments 24 h per day (Kimball, 2005; Ruiz-Vera *et al.*, 2013). Radiometers were calibrated against a black body temperature source before the start of each season to ensure accuracy. During the growing season, sensor lenses were cleaned weekly, and data were inspected daily to ensure that the sensors were within the range of the setpoint. Canopy temperature was recorded at 10 min intervals throughout the season. The dimmer system was programmed to stop heating when canopy temperatures exceeded 40 °C. Ambient reference plots were located adjacent to each pair of heated experimental plots, resulting in two reference plots in 2017 and 2018 and four reference plots in 2019. Accumulated growing degree days (GDDs) were calculated using the equations of Campbell and Norman (1998) as explained by Ruiz-Vera *et al.* (2018).

T-FACE systems increase the vapor pressure gradient from inside the leaf to the atmosphere and increase evapotranspiration (Kimball, 2005). Kimball (2005) suggested that for experiments in which water stress is to have no part, reference plots could be irrigated to meet the increased water requirements for evapotranspiration. In the 3 years of our study, summer (June, July, and August) rainfall was 25–151% greater than the historic normal. Plots were irrigated once on 6 June 2018; however, there was potential that additional irrigation would cause flooding stress, so irrigation was limited to a single time.

Development and pod counts

Vegetative and reproductive developmental stages were scored every 2–4 d within each treatment and cultivar according to the methods of Fehr *et al.* (1971). In 2017, all plants in a 4 m row segment were observed and development was scored based on the majority (>50%) of the plants. Three plants within the rows were also monitored for pod counts on nodes 7–13. In 2018 and 2019, development of five plants within each treatment and cultivar was monitored over the growing season. These same plants were observed for pod counts on nodes 7–9 in 2018 and 10–13 in 2019. The difference between maximum number of pods observed at each node and the number of pods counted at the final harvest was used to estimate the number of aborted pods.

Leaf area index

Leaf area index (LAI) was measured using a LICOR LAI-2000 Leaf Canopy Analyzer (Lincoln, NE, USA) once per week in the early morning or late evening. Measurements were taken along the length of a row following the recommended procedure for row crops in the manual. Two repetitions of one above-canopy reading and four below-canopy readings were taken. Below-canopy readings were distributed evenly between two rows. Due to the proximity of the soybean cultivars and size of the heating arrays, measurements of LAI included both cultivars.

Above-ground biomass and yield measurements

End-of-season AGB, seed yield, and yield components were measured when all plants within a temperature treatment reached maturity (stage R8, Fehr *et al.*, 1971). Harvests were done on 28 September and 18 October 2017, 9 and 22 October 2018, and 8 and 24 October 2019. For each genotype in each control or heat treatment plot, all plants in one row (2 m length) were cut 0.5 cm above the ground, banded, and taken to a field lab for processing. The number of plants harvested from the rows was recorded and ranged from 24 to 32 (average 27). Leaf, stem, and pod biomass were separated, and leaves were discarded. A total of 100 randomly selected pods from each genotype, temperature treatment, and ambient plot were sampled to estimate the average number of seeds per pod. Stems and pods were placed in paper bags and dried at 70 °C for at least 1 week before dry AGB was measured. Seeds were removed from pods and dried for an additional 24 h. Total seed mass and the mass of a random sample of 200 seeds from each treatment and ambient plot were measured. HI was calculated using dry AGB and dry seed mass. The number of seeds produced was estimated using the mass of 200 seeds and the total seed mass. Seed mass and mass of 200 seeds were used to estimate the number of seeds per plant. Yield (kg ha⁻¹) was estimated by multiplying seed mass per plant by planting density.

Statistical analysis

The aim of the study was to determine temperature response functions for soybean developmental, physiological, and yield traits, so

regression analysis was used. Additionally, using the years of the experiment as replicates ($n=3$), ANOVA was performed to determine the effects of soybean cultivar, temperature treatment, and their interaction on soybean development, physiology, and yield. For developmental traits, a second-order polynomial was used to represent the rate of main stem node development with the intercept fixed at the origin (0,0) representing zero nodes on the planting date (PROC REG, SAS 9.4). Models were fit individually for each heat treatment (control, +1.7, +2.6, +3.6, and +4.8 °C). Models combining control data with each of the heat treatments were also created, and an *F*-test was used to determine if individual models fit the data significantly better than the combined model to test for statistically significant differences in the rate of vegetative development between temperature treatments. Regression analysis was performed using the 24 h average canopy temperature through the start of each reproductive stage and 24 h average canopy temperature through the end of the reproductive stage for time to stage and time in stage, respectively (PROC REG; SAS 9.4).

LAI was modeled based on day of year (DOY) [$LAI=y_0+a(DOY)+b(DOY)^2$] using Proc NLIN (SAS 9.4). As for developmental traits, models were fit for each treatment individually and for combined control data with each heat treatment from the same year, and an *F*-test was used to determine significant differences between control and heat treatments. Individual models were used to estimate the maximum LAI value reached during the season.

For yield and yield component traits with a significant ($P<0.05$) effect of cultivar based on ANOVA, regression analysis was performed independently for each cultivar. When the main effect of cultivar was not significant, regression analysis was performed on the combined data. *F*-tests were performed to determine if combined and separate MG models were significantly different. Both linear and quadratic fits were tested to determine which better explained the data, based on R^2 and *P*-values. *F*-tests were performed to determine if linear and quadratic models were significantly different. When quadratic relationships were identified, the axis of symmetry (the line perpendicular to the *x*-axis that passes through the vertex of the curve) was used to identify the maximum *y*-value, indicating threshold temperatures.

Results

Seasonal canopy temperature

All three growing seasons were slightly warmer than the 30 year averages observed from 1980 to 2010 (Supplementary Table S1); 2018 and 2019 were also much wetter (Supplementary Table S1). Ambient seasonal average air temperatures ranged from 22.4 °C (2017) to 23.1 °C (2018), and average canopy temperatures were 0.2–1.5 °C cooler than air temperatures. In 2017 there was the least amount of precipitation during the growing season (June–September) while 2018 had the most, with nearly three times greater rainfall than the 30 year average (Supplementary Table S1). The target increases in canopy temperature of +1.5, +3.0, +4.5, and +6.0 °C resulted in increased canopy temperatures of +1.7, +2.6, +3.6, and +4.8 °C, averaged across the 3 years. Average daily low temperature ranged from 14.7 °C in control plots to 19.6 °C under +6.0 °C treatments. Canopy temperatures were higher in 2018 compared with 2017 and 2019, especially early in the growing season (Supplementary Fig. S2).

Development

Temperature treatments significantly affected the rate of vegetative development (Fig. 1). Both soybean cultivars developed main stem nodes significantly faster than control plants under +1.7 °C and +2.6 °C treatments ($P<0.05$; Fig. 1A, B). Plants grown under the +3.6 °C treatment did not develop stem nodes at a significantly different rate from the control plots ($P>0.05$). Stem node development under the +4.8 °C treatment was significantly accelerated for MG 3.6 but was significantly slowed for MG 4.2 ($P<0.05$; Fig. 1A, B) based on days after planting. It took more GDDs to reach a given vegetative developmental stage with increasing temperature treatments (Fig. 1C, D). This trend was true for both soybean cultivars.

The timing and duration of reproductive stages when measured as days or thermal time did not differ between MG 3.6 and MG 4.2 (Table 2). The time to the start of reproductive stages R1–R3 was measured as days after planting and was not significantly correlated with average canopy temperature ($P>0.05$; Fig. 2A–C). Onset of later reproductive stages (R4–R6) was correlated with temperature and slightly delayed in higher temperature treatments (Fig. 2D–F). The duration of individual reproductive growth stages in days was not significantly altered by growth temperature (Table 2), and all plots reached R6 within 1 week of each other. The 2018 planting date was 2 weeks earlier than in 2017 and 2019, and plants reached R6 ~1 week later in 2018 compared with 2017. In thermal time, the start of stages R1–R6 was significantly correlated with average canopy temperature ($P<0.05$; Fig. 2G–L).

Leaf area index

LAI developed more slowly and peaked at lower maximum values under the higher heat treatments (+3.6 °C and +4.8 °C) in all years of the study ($P<0.05$, Fig. 3A–C). Milder heat treatments (+1.7 °C and +2.6 °C) were significantly different from control plots in 2017 and 2019 ($P<0.05$), but not in 2018 ($P>0.05$). There was a significant linear decline in maximum LAI with increasing average canopy temperature, and peak LAI decreased by ~2 units between ambient and +4.8 °C (Fig. 3D).

Table 2. ANOVA (F , P) of the effects of temperature treatment (Temp), cultivar (Cv), and their interaction on the maximum number of pods initiated, the number of aborted pods, and the final pod number on nodes 7–13

	Temp	Cv	Temp×Cv
Maximum pods per node	0.77, 0.5494	0.18, 0.6701	1.49, 0.2235
Aborted pods per node	0.53, 0.7164	0.20, 0.6567	1.31, 0.2851
Final pods per node	4.75, 0.0033	0.11, 0.7431	0.39, 0.8145

Bold indicates statistical significance ($P<0.05$).

Pod number, above-ground biomass, and yield

The maximum number of pods initiated per node and the final number of mature pods were negatively correlated with 24 h average canopy temperature during reproductive development (Fig. 4A, C). Canopy temperature was not significantly correlated with the number of aborted pods (Fig. 4B). Cultivars did not differ in the maximum number of pods, number of aborted pods, or final number of pods produced on nodes 7–13 (Table 2).

The two soybean cultivars significantly differed in seed yield, HI, and individual seed mass (Table 3). MG 3.6 had greater yield, HI, and seed size compared with MG 4.2 (Fig. 5A, C, D). However, the cultivars did not have significantly different ABG, seed number per pod, or seed number per plant (Table 3; Fig. 5B, E, F). Quadratic models better explained the temperature response of yield, HI, seeds per pod, and seeds per plant than simple linear models (Supplementary Table S2). Yield decreased in both MG 3.6 and 4.2 when season average canopy temperatures exceeded 20.9 °C and 22.7 °C, respectively. AGB per plant was not significantly affected by canopy temperature (Fig. 5B; Supplementary Table S2). HI in both soybean cultivars showed a significant quadratic relationship with season average canopy temperature and decreased when mean season temperature exceeded 22 °C (Fig. 5C; Supplementary Table S2).

Decreased seed yield with increasing temperature resulted from a combination of decreased individual seed mass, fewer seeds per pod, and fewer seeds per plant (Fig. 5D–F). Individual seed mass was negatively correlated with temperature in both MG 3.6 and MG 4.2, decreasing by 0.5 g and 0.8 g per °C, respectively (Fig. 5D). The number of seeds per pod and seeds per plant showed quadratic relationships with temperature (Fig. 5E, F; Supplementary Table S2). The number of seeds per pod decreased when season mean temperature exceeded 22.0 °C (Fig. 5E), and seeds per plant decreased when season mean temperature exceeded 22.8 °C (Fig. 5F).

Discussion

This study examined soybean responses to season-long increases in temperatures across three growing seasons, aiming to identify temperature thresholds for yield loss. We hypothesized that soybean yields would respond non-linearly to increasing temperature, with accelerated development and damage to reproductive structures contributing to yield loss. Results supported the first hypothesis. Yield showed a non-linear response to temperature, with decreases apparent when canopy temperature exceeded 20.9 °C for the locally adapted cultivar (MG 3.6) and 22.7 °C for a cultivar adapted to more southern locations (MG 4.2). The seasonal average temperature of even the mildest elevated temperature treatment

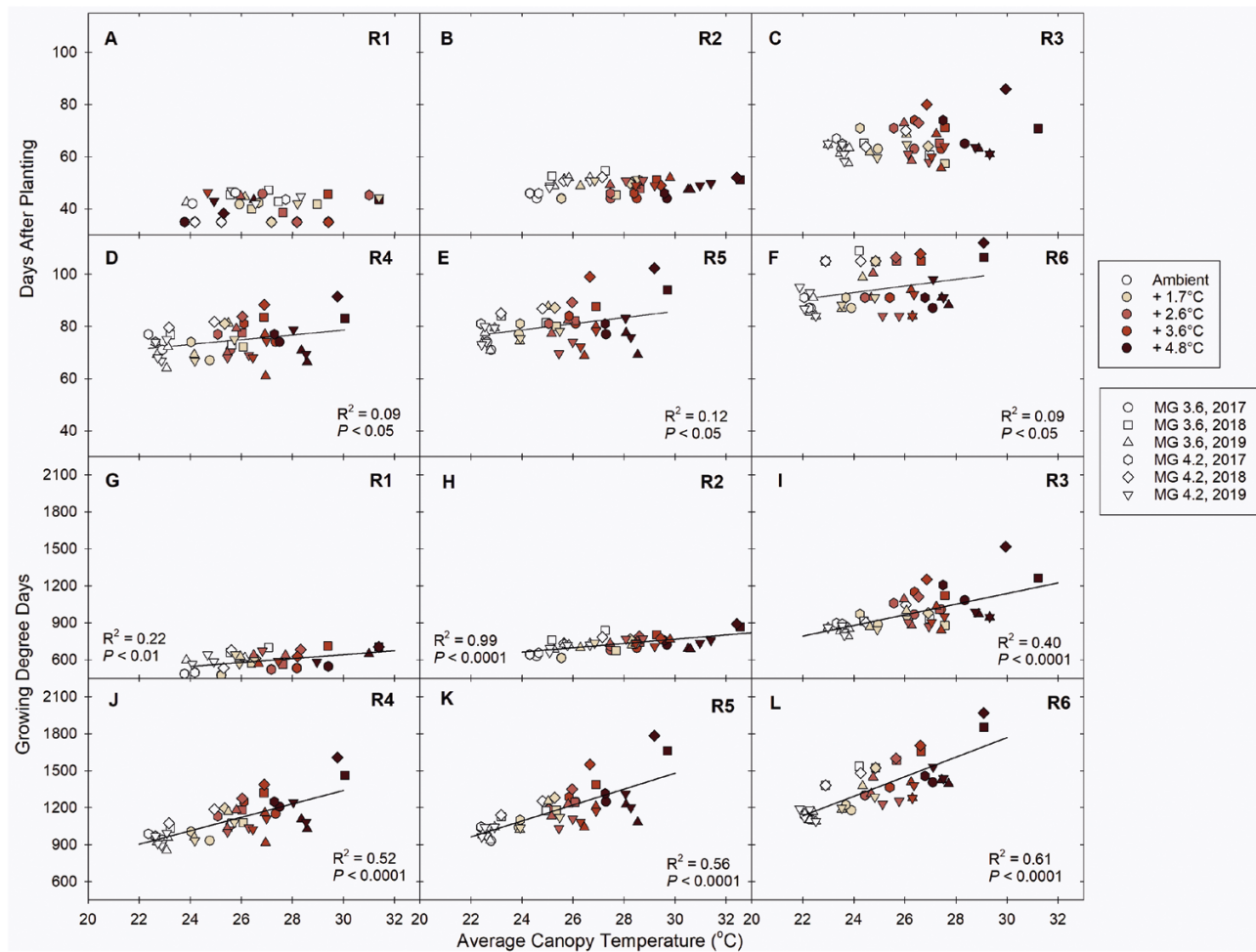


Fig. 2. Relationships between days after planting or growing degree days and the average canopy temperature at the start of each reproductive stage. Regression lines and statistics are shown for significant linear relationships ($P < 0.05$). R1, beginning bloom; R2, full bloom; R3, beginning pod; R4, full pod; R5, beginning seed; R6, full seed.

exceeded these threshold temperatures, which reinforces that any further increase in global temperature from climate change will negatively impact soybean productivity in this region of the US Corn Belt. The results did not necessarily support the hypothesis that acceleration of development at high temperatures would contribute to yield loss. Elevated temperature accelerated progression through vegetative development, resulting in more main stem nodes, but did not alter reproductive developmental stages. Elevated temperatures had a pronounced effect on LAI, with the highest temperature treatment (+4.8 °C) reducing maximum LAI from ~6.5 to 5.0. Elevated temperature decreased production of pods, resulting in fewer pods and seeds, and HI decreased non-linearly as growing temperatures exceeded 22 °C. The similarity in the HI and seed yield responses supports a previous suggestion that HI is a valuable characteristic for screening soybean varieties for heat tolerance (Boote *et al.*, 2018). These results have implications for modeling the effects of temper-

ature stress on soybean productivity, suggesting that effects on LAI and pod and seed production are key determinants of yield loss, and not reduced leaf-level carbon assimilation (Kumagai *et al.*, 2022).

One caveat to our results is that the infrared heaters increase vapor pressure deficit (VPD) as well as temperature. However, infrared heaters have advantages over chambers or tunnels in that natural air flow is maintained and shading is avoided (Kimball, 2005). Infrared heaters also warm the canopy, which is similar to solar heating of leaves, and the system is more energetically efficient than heating air because of boundary layer resistance (Kimball, 2005). Adding water to counteract the VPD differences is one strategy to counterbalance the VPD problem, and we irrigated as the seasons allowed, although the seasons were all significantly wetter than normal (Supplementary Table S1). Additionally, previous experiments using the T-FACE system did not find any difference in leaf hydraulic conductance of soybeans grown at +3.5 °C (Locke *et al.*, 2013). While we cannot eliminate

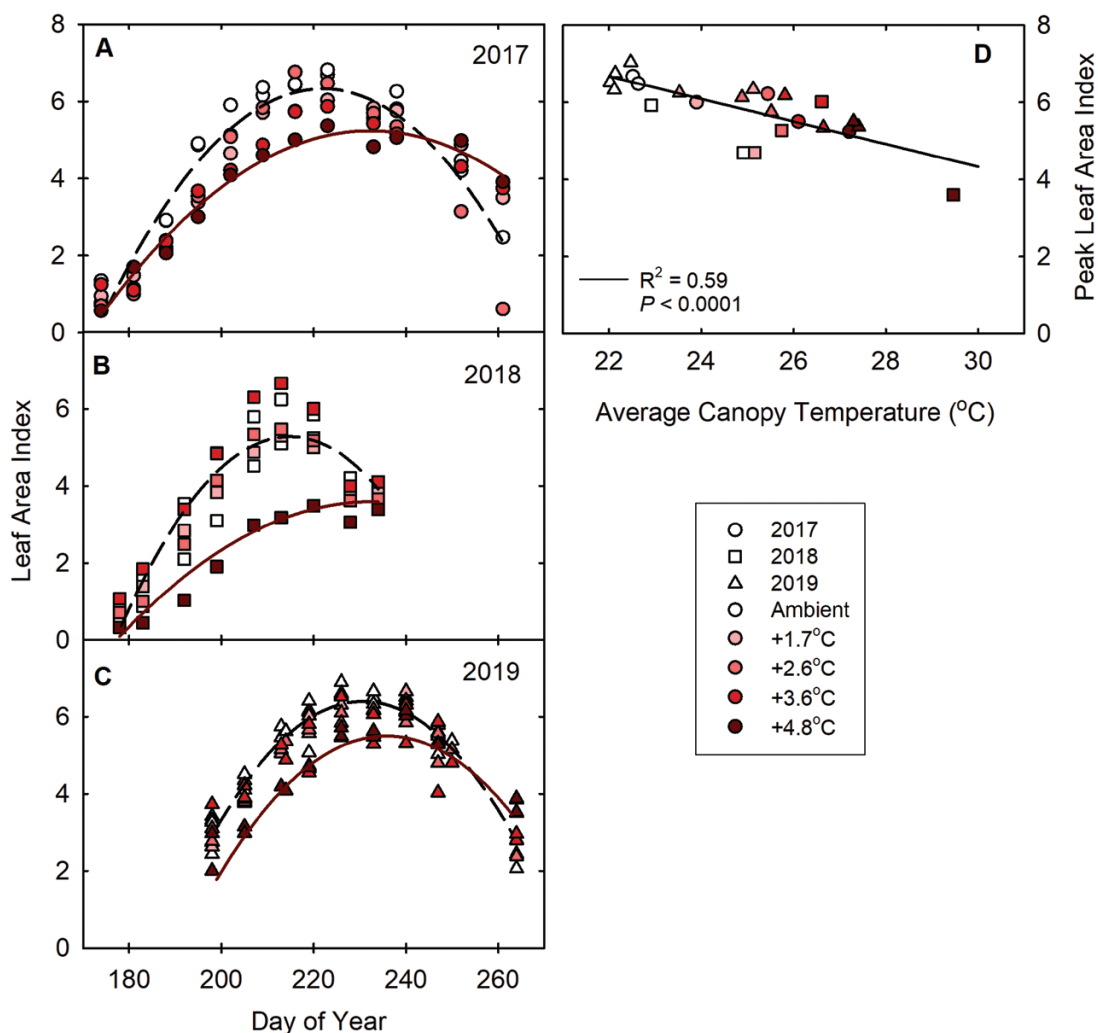


Fig. 3. Leaf area index (LAI) measured in ambient and four elevated temperature plots in 2017 (A), 2018 (B), and 2019 (C). Dashed lines indicate the non-linear fit for ambient LAI, and solid lines show the fit for the +4.8 °C treatment. (D) The correlation between maximum leaf area index and seasonal average canopy temperature across 3 years and five different temperature treatments.

the possibility that VPD contributed to the responses that were measured, we did ensure that plants were not water limited.

Elevated temperature increased vegetative nodes, but decreased leaf area index

Moderate increases in canopy temperature accelerated vegetative development, resulting in more main stem nodes (Fig. 1). Previous research of a different soybean cultivar grown at elevated temperature (+3.5 °C) reported that one additional main stem node was formed in plants grown at high temperature (Ruiz-Vera et al., 2018). However, studies of an older cultivar (Thorne) reported inconsistent effects of temperature on the number of main stem nodes (Köhler et al., 2017). A controlled-environment study investigating the independent effects of elevated temperature during vegetative and reproductive growth found that elevated temperature increased main stem node number when

applied during vegetative growth, but not during reproductive growth (Allen et al., 2018). More main stem nodes did not support greater leaf area, and LAI declined linearly with increasing temperature. Additional stem nodes provide additional sites for leaf development, potentially supporting greater leaf area (Allen et al., 2018), but our results found the opposite. LAI peaked in the ambient treatment at ~6.7 and declined by 0.3 LAI units per °C increase in temperature (Fig. 3). The leaf expansion rate in soybean was previously modeled to be greatest at an air temperature of ~26 °C (Setiyono et al., 2007). The ambient canopy temperature in our study was ~22 °C from the start of the growing season up to peak LAI, and our results suggest that while temperatures above 22 °C accelerated addition of main stem nodes, higher temperatures decreased individual leaf expansion rates resulting in lower LAI. Our study included two indeterminate soybean genotypes and our estimates of LAI included both of those because of the limited size of the elevated temperature plots. We

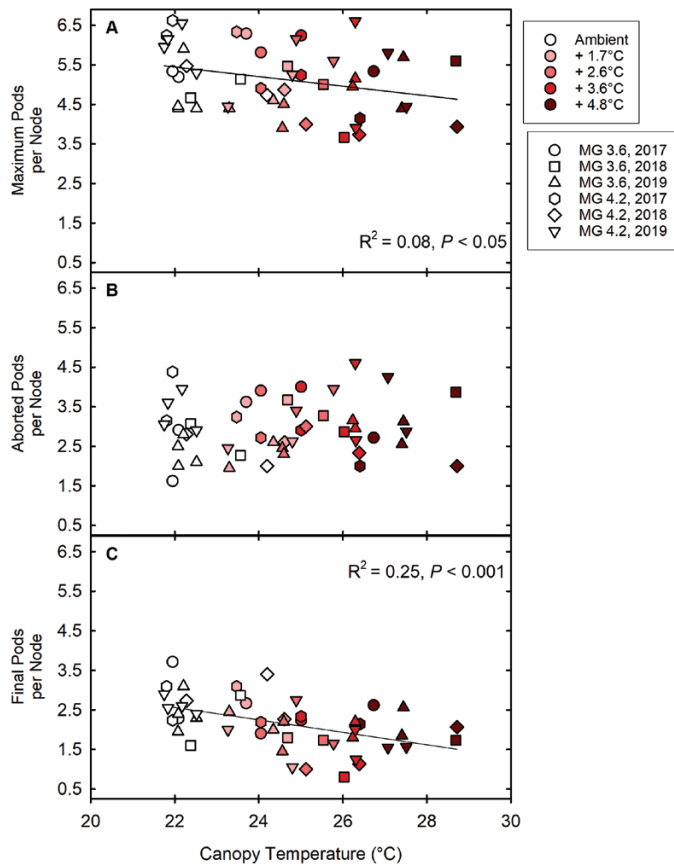


Fig. 4. Correlation between the maximum number of pods initiated per node (A), the number of aborted pods per node (B), and the final number of pods per node (C) and canopy temperature measured during reproductive growth (start of R1 through end of R5). Lines indicate significant correlations. Pod counts were collected from nodes 7–13.

Table 3. ANOVA (*F*, *P*) of the effects of temperature treatment (Temp), soybean cultivar (Cv), and their interaction on yield component traits

	Temp	Cv	Temp×Cv
Yield (kg ha ⁻¹)	5.09, 0.0030	9.31, 0.0047	0.92, 0.4637
AGB (g)	1.60, 0.2010	3.30, 0.0791	0.74, 0.5699
HI	5.31, 0.0023	6.58, 0.0156	0.23, 0.9220
200 seed mass (g)	5.35, 0.0024	51.87, < 0.0001	0.24, 0.9129
Seeds per pod	3.70, 0.0145	1.49, 0.2324	0.48, 0.7504

HI, harvest index; AGB, above-ground biomass. Bold indicates statistical significance (*P* < 0.05).

found that LAI was lower at the start of each of the later reproductive stages (R4–R6) in the elevated temperature treatments (Fig. 4), despite peak LAI occurring later in the growing season at higher temperatures (Fig. 4). Maximum leaflet size on a soybean plant increases from lower nodes to the middle canopy nodes (~6–8), then decreases on upper canopy nodes (Setiyono *et al.*, 2011). Additionally, leaf drop is more rapid in nodes initiated before flowering (Setiyono *et al.*, 2011).

Elevated temperature delayed the start, but not the duration, of reproductive stages

Previous research reported that progression through reproductive development stages slows at average growing temperatures above 23 °C (Thomas *et al.*, 2010; Choi *et al.*, 2016; Boote *et al.*, 2018). We found that the start of reproductive developmental phases was increasingly delayed in later stages (R4–R6) as temperature increased (Fig. 2), but the length of each reproductive stage was not altered by temperature in contrast to our hypothesis and other prior work (Choi *et al.*, 2016). All plots from the same growing season reached R6 within 7 d (Table 1). Meanwhile, there was a clear difference in development time between different seasons (Table 1), with time to the start of R6 in 2018 lasting 9–21 d longer than in 2017 and 2019. Progression through reproductive development is affected by both temperature and photoperiod (Garner and Allard 1930; Major *et al.*, 1975; Setiyono *et al.*, 2007; Archontoulis *et al.*, 2014). Long photoperiods delay floral initiation and increase the duration of post-flowering phases (Thomas and Raper, 1976; Hadley *et al.*, 1984; Caffaro and Nakayama, 1988; Summerfield *et al.*, 1998). The planting date in 2018 was ~2 weeks earlier than in 2017 and 2019. Variation in planting date can cause the time to flowering to double in some soybean cultivars (Sinclair *et al.*, 1991). Consequently, longer photoperiods early in growth and development may have delayed the start of R6 in 2018.

The relationship between accumulated thermal energy and rate of development changed between plots at different growing temperatures. During each season, the number of GDDs accumulated before the end of seed fill (stage R6) increased with increasing temperature treatment. Crop development models that use cumulative heat exposure predict that, under these conditions, the experimental plants would develop faster in terms of real days, while maintaining a consistent number of GDDs (McMaster and Wilhelm, 1997). Instead, development in real time either progressed at the same rate or was slightly delayed. In terms of thermal time, the rate of vegetative development slowed, and reproductive stages began later in the season while lasting longer as growing temperature increased. Previous studies of soybean development under different temperature conditions also found that the relationship between development and thermal time changed with growing temperature as plants accumulated more GDDs before the start of vegetative and reproductive stages (Heinemann *et al.*, 2006; Ruiz-Vera *et al.*, 2018). These results support the conclusion that the GDD model becomes less reliable as a predictor of crop development at supra-optimal growing temperatures.

Yield components decline with increasing temperature

Yields were significantly reduced by increasing growing season temperature, and both the maximum and final number of pods produced on main stem nodes were negatively correlated with canopy temperature (Fig. 4). This suggests that both flower and

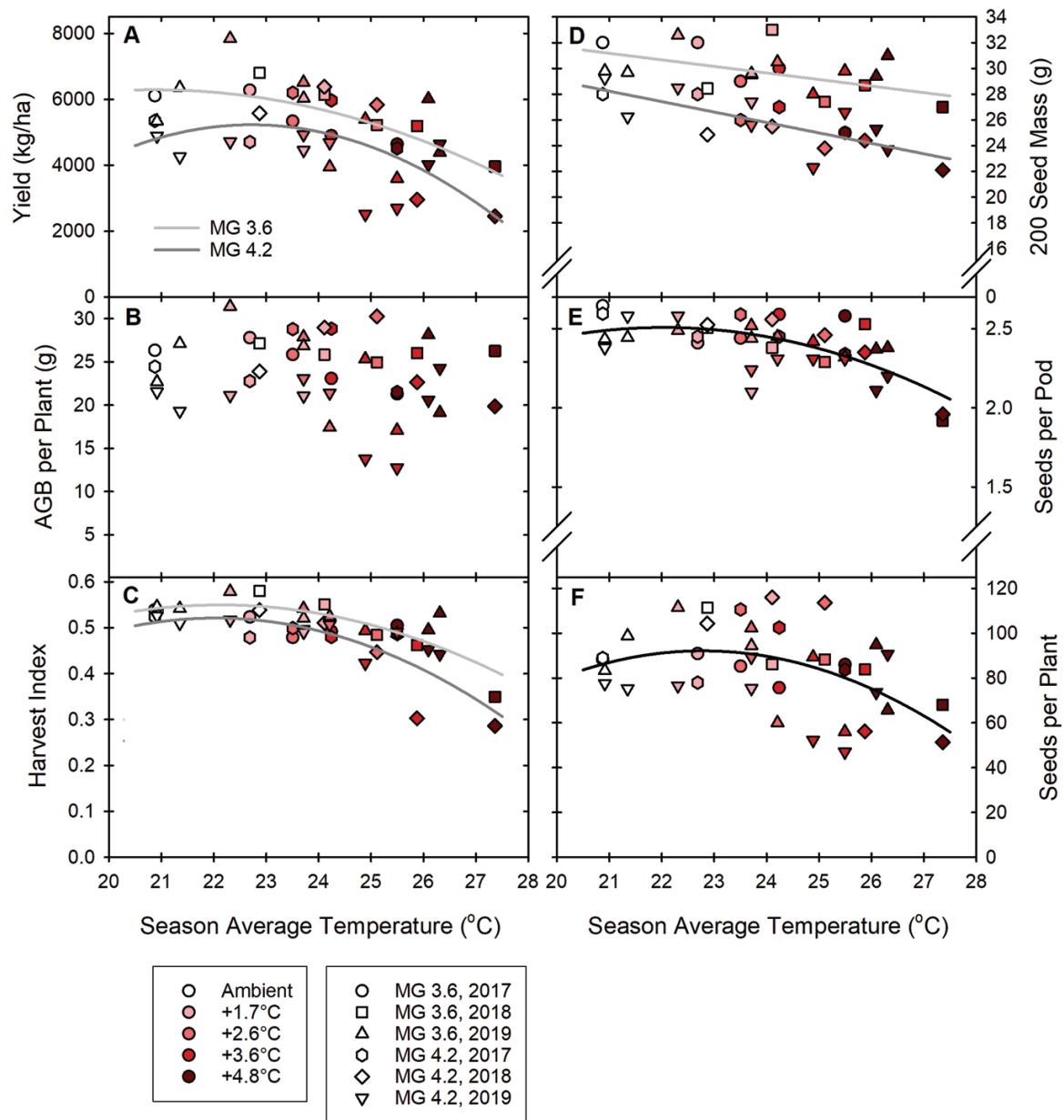


Fig. 5. Correlation between (A) seed yield (kg ha^{-1}), (B) above-ground biomass (g), (C) harvest index, (D) individual seed mass (g), (E) seed number per pod, and (F) seed number per plant and growing season average canopy temperature for two cultivars of soybean grown in three different growing seasons. Regression analysis was done separately for each cultivar when there was a significant main effect of cultivar for that trait (light gray curve, MG 3.6; dark gray curve, MG 4.2) ($P < 0.05$). Otherwise, data from both cultivars were combined for analysis (black curve).

pod production were sensitive to temperature, but abortion of pods was not temperature sensitive (Fig. 4). The percentage of pod set in soybean reportedly varies from 18% to 46% (Gai et al., 1984), and our estimates of pod set were within this range. Pod number in soybean has been modeled as a function of assimilate supply (Egli, 2017); however, timing of pod initiation also determines survival, with early initiated pods at a given node surviving longer than pods initiated later (Egli and Bruening, 2002). The delay to the start of the reproductive period combined with

lower LAI may have both contributed to fewer pods at elevated temperatures, which was probably a major contributor to yield loss. A prior study of soybean grown under greenhouse conditions reported that pod number decreased at average air temperatures above 27.5 °C, and pod number was a major determinant of seed yield through effects on seed number (Choi et al., 2016). Lin et al. (2020) reported that increasing night temperature above 22 °C in a temperature gradient chamber had a negative effect on pod set in soybean, in one of the 2 years of study.

Lower temperature thresholds for yield decreases in our study compared with others may be caused by harsher conditions in the field compared with more controlled environments. Average ambient temperatures in this experiment were <23.5 °C and average night temperatures were <19 °C. Under these conditions, even mild heat treatments (+1.7 °C) tended to reduce pod numbers and seed yield per plant. Allen *et al.* (2018) proposed that higher growing temperatures could increase soybean yields by increasing the number of stem nodes and, subsequently, the number of pods per plant. Soybean yield per plant is heavily dependent on the number of reproductive stem nodes (Kahlon *et al.*, 2011). In our study, higher growing temperatures increased stem development, but also reduced final mature pods per node, seeds per pod, seed mass, and yield per plant. This outcome is likely to be due to the sensitivity of flower, pod, and seed development to heat stress relative to stem node development (Egli and Wardlaw, 1980; Sage *et al.*, 2015). For example, heat wave experiments exposing soybeans to elevated (+5.0 °C and +6.0 °C) temperatures for a few days at a time during reproductive development reported significantly reduced pods per plant and seed yield, despite no evidence of long-term effects on physiology or vegetative development (Siebers *et al.*, 2015). In our temperature gradient study, fewer initiated pods, fewer mature pods, reduced seed size, and fewer seeds per pod indicate that impaired flower, pod, and seed development all contributed to reduced partitioning efficiency and yield.

It has been suggested that the effects of heat stress on yield could be mitigated by growing soybean varieties suited for warmer climates (Kumagai and Sameshima, 2014); however, those varieties would need to be planted earlier in the spring to take advantage of the benefit of a longer growing season. In all 3 years of our study, MG 3.6 and MG 4.2 were planted on the same date in mid to late May, which may have prevented potential differences between the cultivars from being fully realized (Zhang *et al.*, 2007). Further experiments with altered planting dates could test this more fully (Mourtzinis and Conley, 2017).

Conclusions

Ambient growing conditions in the middle of the US Corn Belt are probably at or near the optimum temperature for soybean yields since even the mildest increase in temperature in experimental plots produced consistently lower yields than in the control plots. This decline in yield was driven by changes in LAI, pod number per node, seed size, and HI. These field experiments provide critical data for calibrating and validating models of soybean response to climate change. They also reinforce conclusions drawn from historical data that higher growing season temperatures result in lower soybean yields, even in temperate environments (Kucharik and Serbin, 2008; Mourtzinis *et al.*, 2015; Kukul and Irmak, 2018; Eck *et al.*, 2020).

Supplementary data

The following supplementary data are available at [JXB online](#).

Table S1. Monthly temperature and precipitation data measured at the field site.

Table S2. Linear and non-linear regression analysis of soybean yield component responses to temperature.

Fig. S1. Aerial image of one of the two replicate T-FACE plots in 2019.

Fig. S2. Daily temperature measured in ambient and elevated temperature treatments.

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Author contributions

CHB, CJB, KG, and EAA: conceptualization; KG, CJB, and EAA: funding acquisition; TP and CMM: assisting with temperature treatments; CHB, CAM, EAA, NM, and AMM: performing the experiments and data collection; BP and HK: contributing to data analysis; AEL: assisting with statistical analysis; CHB and EAA: writing the original draft. All authors contributed to interpretation and discussion, and have revised and approved the final manuscript.

Conflict of interest

The authors declare that they have no conflict of interest.

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Data availability

The data supporting the findings of this study are available from the corresponding author, Elizabeth Ainsworth, upon request.

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